Engineering a Resilient Enterprise Windows Infrastructure

By: Ritvik Indupuri

Date: April 25, 2026

1. Executive Summary

This initiative involved the end-to-end architecture and deployment of a secure, centralized Windows domain environment under the `group13.c24200.cit.lcl` namespace. The infrastructure addresses three core enterprise requirements: robust identity governance, high data availability, and automated compliance through centralized patch management. By integrating Active Directory Domain Services (AD DS), Distributed File System (DFS), and Windows Server Update Services (WSUS), a scalable ecosystem was engineered that prioritizes the Principle of Least Privilege (PoLP) and multi-site system redundancy.

2. Professional Technical Stack

- Directory Services: Multi-tier AD DS architecture with delegated `Organizational Unit (OU)` management and inheritance control.
- Security Framework: Role-Based Access Control (RBAC) enforced via scoped group memberships and hardened Access Control Lists (ACLs).
- High Availability: Site-aware DFS Namespaces with multi-target synchronization across the `LAN1913` site.
- Systems Management: `WinRM-based remote automation` and `PowerShell-driven service orchestration` for fleet-wide maintenance.
- Lifecycle Management: WSUS-driven patch staging, synchronization validation, and targeted deployment (OS Build: 10.0.26100.1742).

3. Global Architecture & Logic

The foundational environment is anchored by a centralized Domain Controller acting as the single source of truth for identity and policy enforcement. The architectural strategy separates the core authentication layer from the resource and management layers to ensure modularity and security.

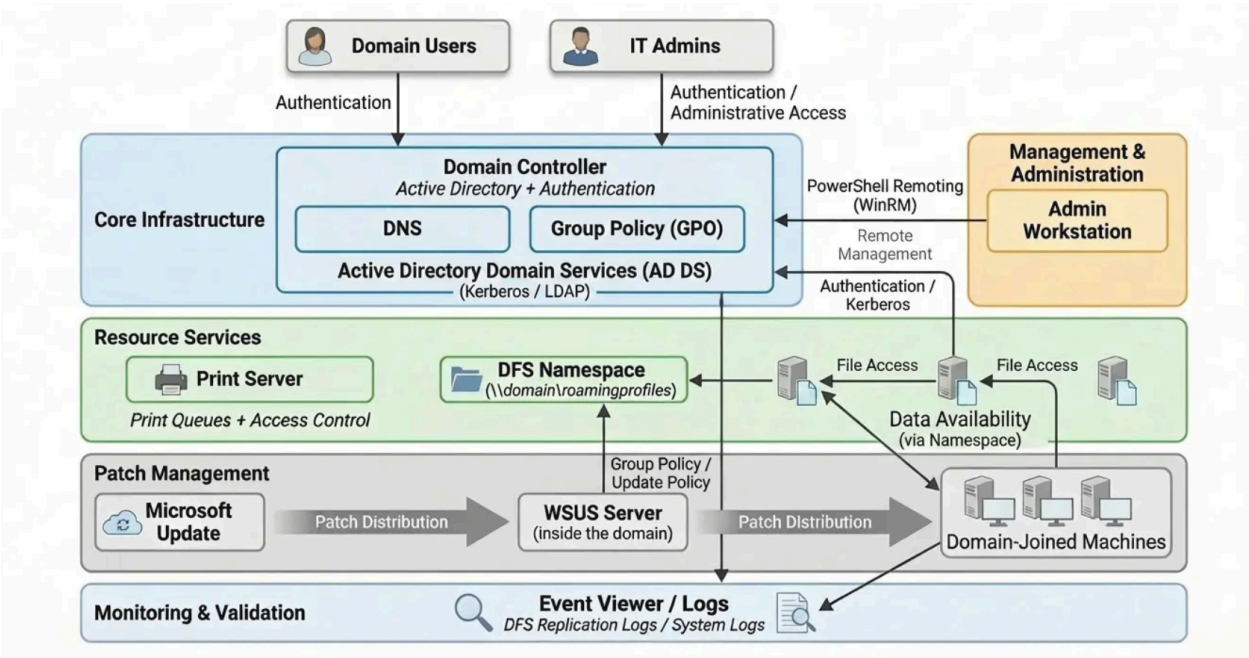
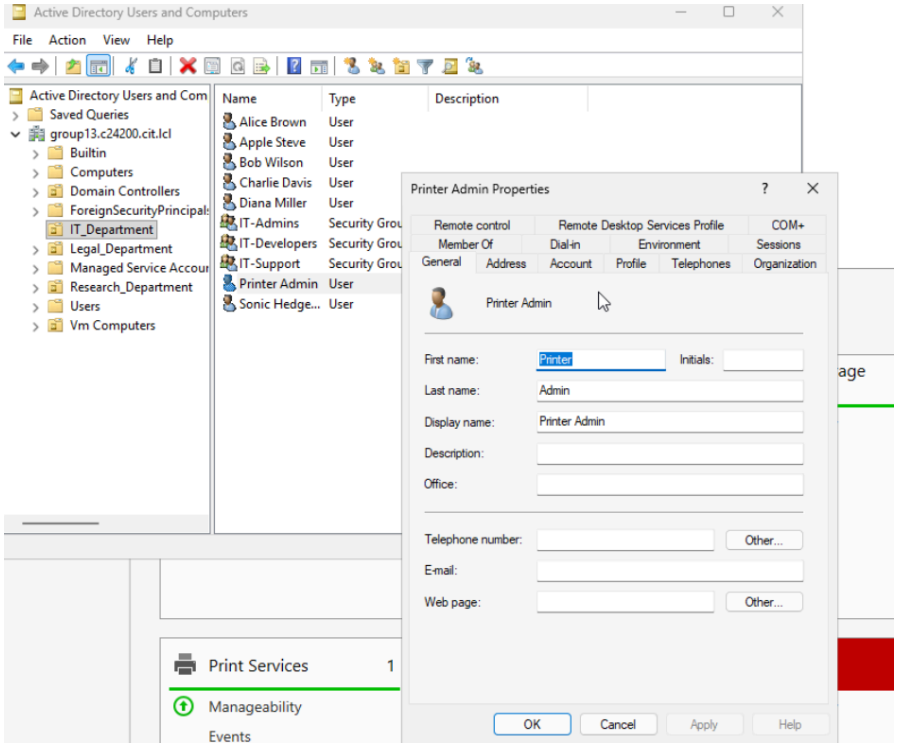


Figure 1: High-level architectural blueprint integrating AD DS, DFS redundancy, and centralized update management.

This design facilitates a Zero-Trust approach to administration. By leveraging secure management workstations and WinRM-encrypted traffic, the requirement for direct interactive logins on sensitive production servers is eliminated, significantly reducing the risk of lateral movement and credential harvesting. All service interactions are brokered through this centralized authentication model to maintain a strict audit trail.

4. Identity Governance & Role Delegation

A primary objective was the implementation of a mature Identity and Access Management (IAM) strategy. Within the `IT_Department` OU, a clear separation between standard user objects and administrative accounts is established. This logical partitioning prevents privilege escalation and ensures that administrative roles are clearly defined within the directory hierarchy.



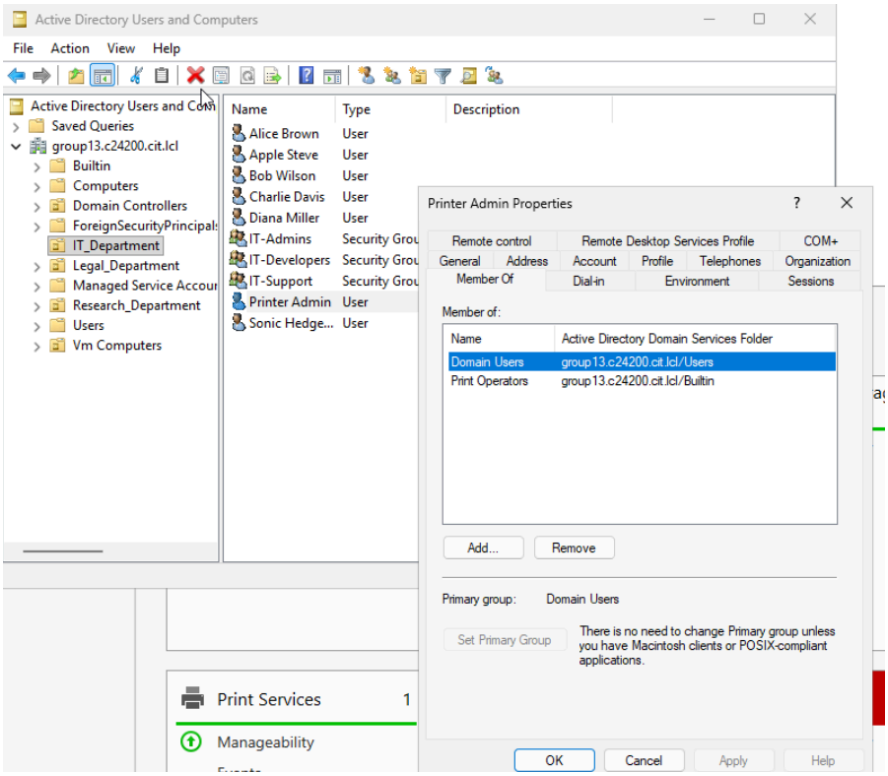
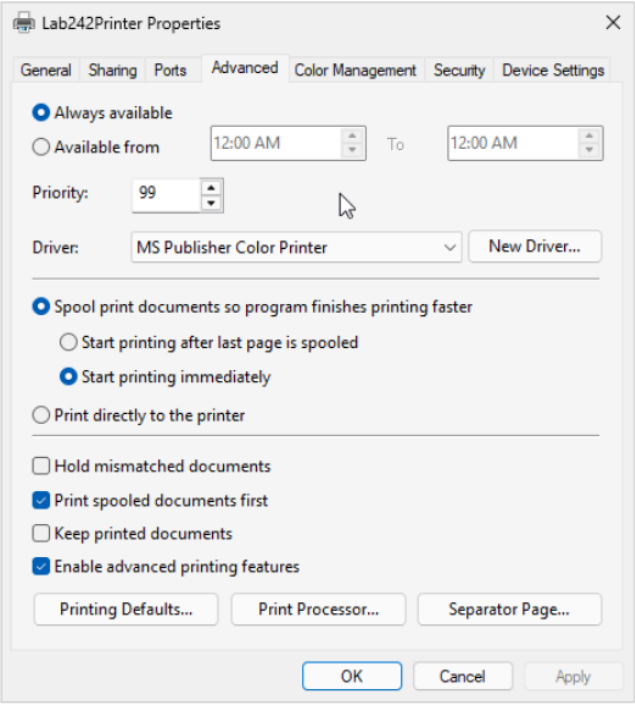


Figure 2: Administrative object properties and RBAC implementation via the Print Operators group membership.

To minimize the blast radius of a potential credential compromise, Role-Based Access Control (RBAC) is utilized. Instead of assigning broad domain administrative rights, the `Printer Admin` user is mapped specifically to the `Print Operators` group. Membership is confined to the built-in `Print Operators` and `Domain Users` containers, ensuring the account possesses only the specific permissions required to manage the print fleet.

5. Resource Management: Enterprise Print Services

Resource governance on the `G13SRV01VM` print server is optimized for both performance and granular security. Hardware is abstracted into managed queues, providing a consistent and secure output environment for all domain-joined clients. The configuration targets both throughput efficiency and strict access control.



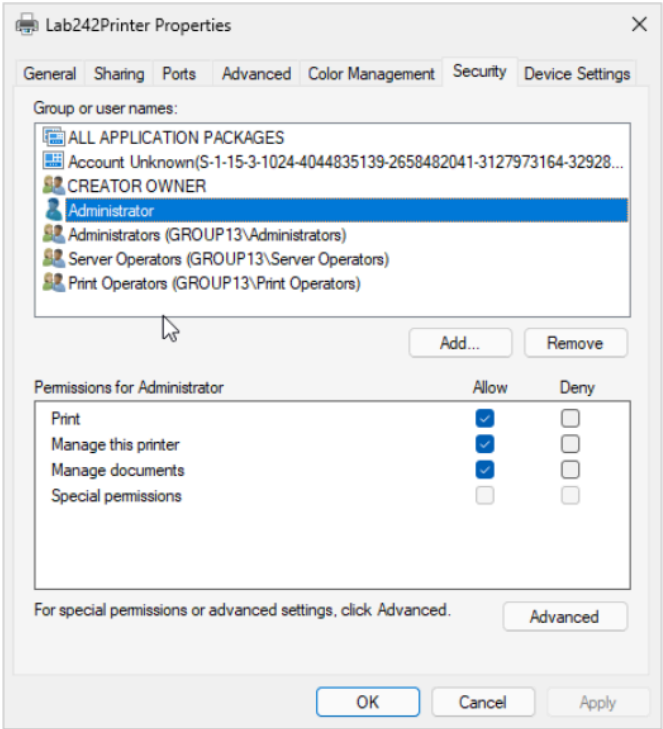


Figure 3: Configuration of high-priority queues (99), hardened security descriptors, and standardized output defaults.

To manage high-volume contention, the `Lab242Printer` is configured with a Priority of 99, ensuring critical workloads are processed first. Access Control Lists (ACLs) are explicitly hardened, granting "Manage documents" and "Manage printer" rights only to the `Print Operators` and `Server Operators` groups. Furthermore, standardizing defaults such as portrait orientation and duplex settings ensures a predictable user experience and efficient resource utilization across the workstation fleet.

6. Remote Administration & Automation Workflow

Enterprise management requires the ability to orchestrate services at scale without physical server presence. A robust remote management framework is implemented using Windows Remote Management (WinRM) and the WS-Management protocol.

```
[G13SRV05]: PS C:\Users\Administrator.GROUP13\Documents> Get-Service spooler

Status   Name               DisplayName
------   ----               -----------
Running  spooler            Print Spooler
```

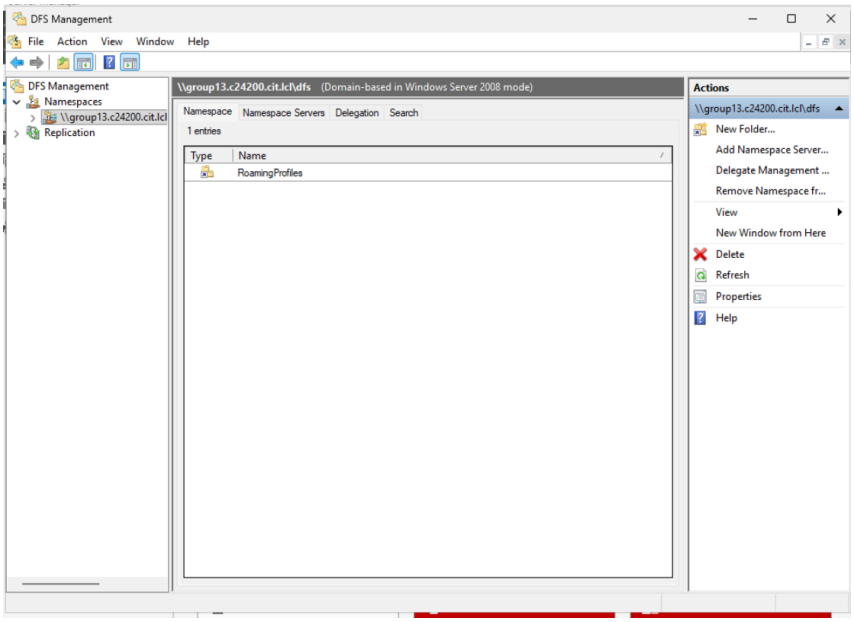
```
[G13SRV05]: PS C:\Users\Administrator.GROUP13\Documents> Get-Service winrm

Status   Name               DisplayName
------   ----               -----------
Running  winrm              Windows Remote Management (WS-Manag...


[G13SRV05]: PS C:\Users\Administrator.GROUP13\Documents>
```

Figure 4: WinRM security validation and PowerShell-based service health verification on remote node G13SRV05.

Using PowerShell Remoting, the health of critical services on remote nodes such as `G13SRV05` is validated. This workflow allows for the rapid identification and remediation of service interruptions, such as Print Spooler failures, without the overhead and security risks of RDP. All management traffic is restricted to validated `TrustedHosts`, ensuring that only authorized administrative nodes can initiate remote sessions.

7. Distributed File Services (DFS) & Data Resilience

To ensure business continuity, a Distributed File System (DFS) namespace is implemented. This layer abstracts physical server paths, providing users with a single, highly-available logical path to roaming profiles and sensitive data. The topology is designed to survive individual server failures without impacting end-user access.



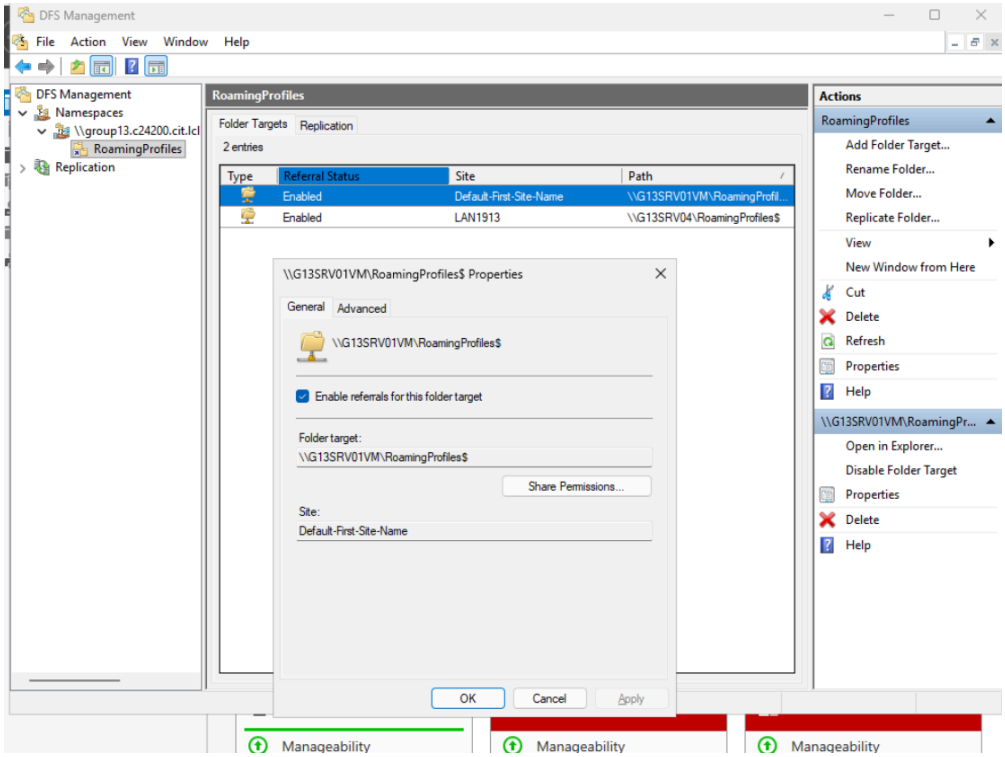
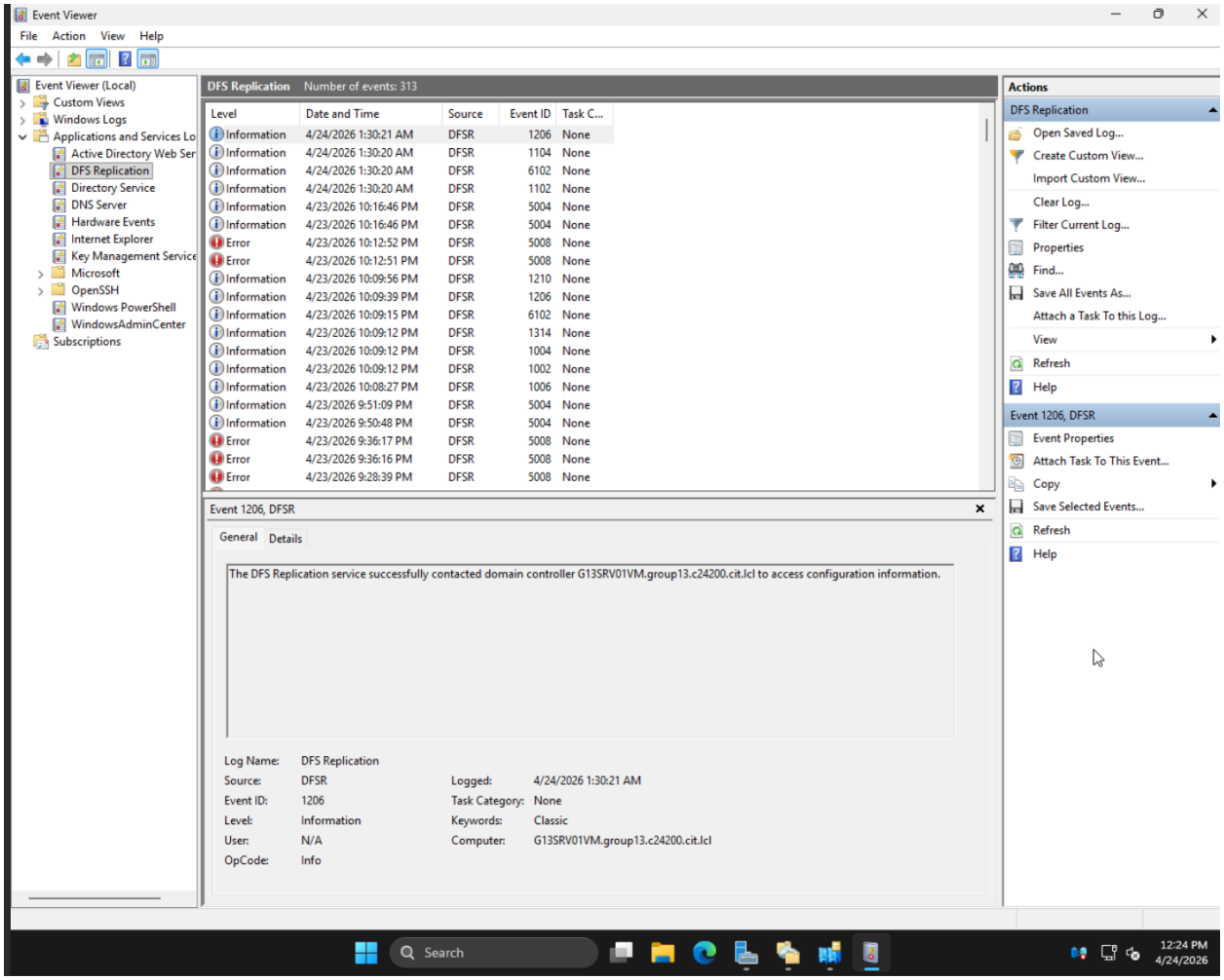


Figure 5: Logical namespace management and multi-target referral configuration supporting the LAN1913 site.

The namespace targets, running in Windows Server 2008 mode, are synchronized between `G13SRV01VM` and `G13SRV04`. By enabling site-aware referrals, the infrastructure intelligently directs client traffic to the nearest available server, optimizing bandwidth and ensuring uninterrupted access during server maintenance cycles. This abstraction ensures that storage migration or hardware refreshes can occur transparently to the user base.

8. Monitoring, Validation, and Log Analysis

Continuous health validation is achieved through a combination of replication proofs and rigorous event log analysis. This ensures that high-availability features are not just configured, but actively functioning within acceptable parameters. Proactive monitoring reduces the mean time to recovery (MTTR) for critical data services.



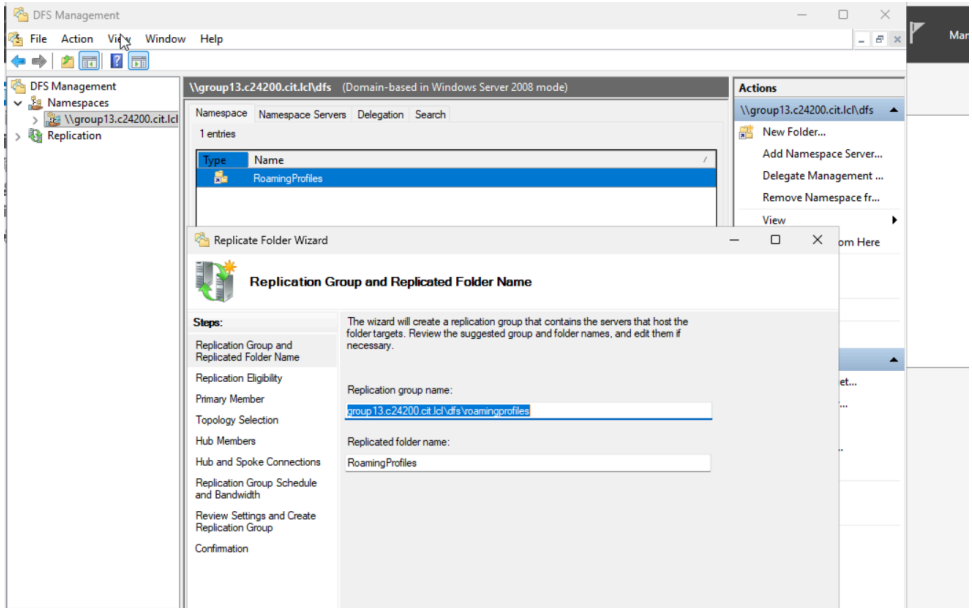
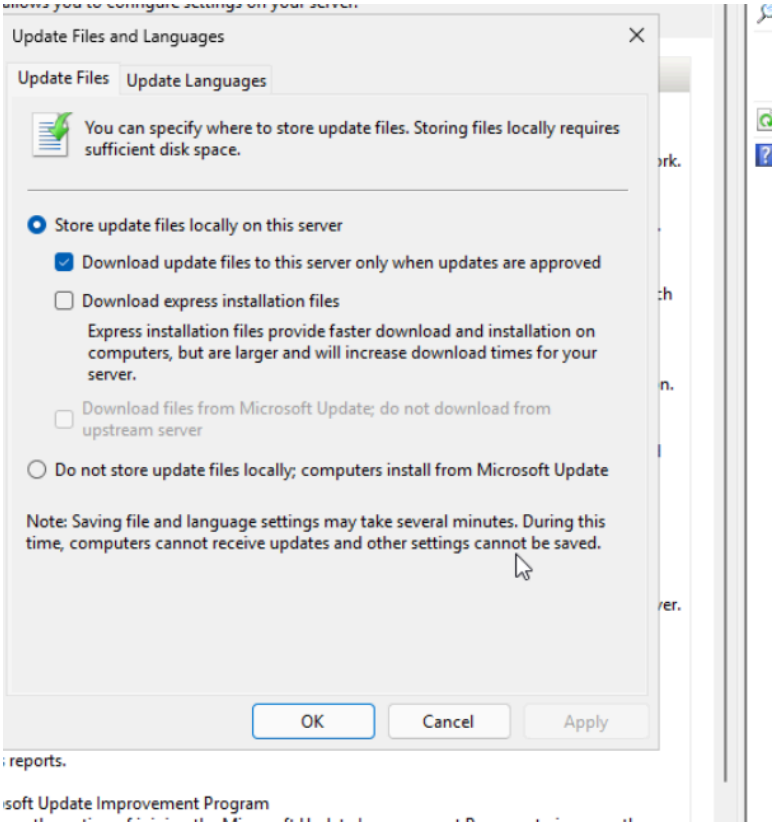


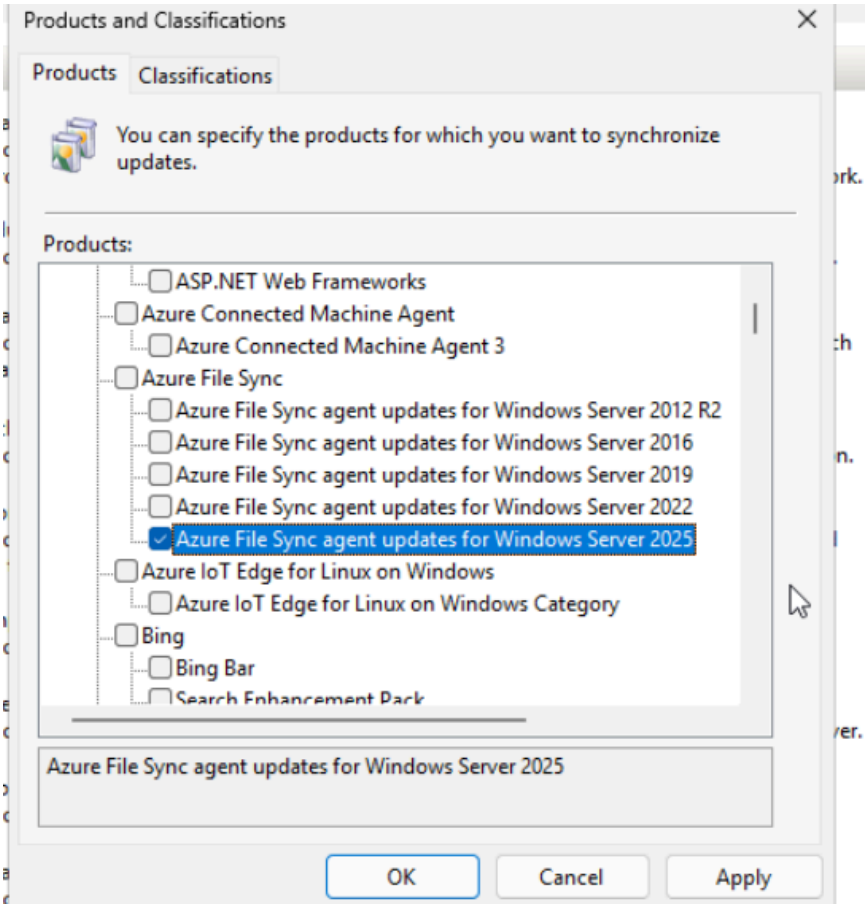
Figure 6: Replication group verification and diagnostic event log analysis (Event ID 1206).

Analysis of `Event ID 1206` confirms that the DFSR service successfully negotiated configuration updates with the domain controller (`G13SRV01VM.group13.c24200.cit.lcl`). This diagnostic step is a critical component of professional system validation, providing technical proof that the multi-server synchronization engine is operating in a healthy, synchronized state across all replication partners.

9. Managed Patch Lifecycle (WSUS)

To maintain a secure security baseline, a centralized WSUS infrastructure is deployed on `G13SRV07`. This ensures that all domain assets receive validated and approved security updates according to a controlled, audited schedule, preventing the deployment of unstable or unauthorized patches.


Update Files and Languages
Update Files
Update Languages
You can specify where to store update files. Storing files locally requires sufficient disk space.
Store update files locally on this server
Download update files to this server only when updates are approved
Download express installation files
Express installation files provide faster download and installation on computers, but are larger and will increase download times for your server.
Download files from Microsoft Update; do not download from upstream server
Do not store update files locally; computers install from Microsoft Update
Note: Saving file and language settings may take several minutes. During this time, computers cannot receive updates and other settings cannot be saved.
OK
Cancel
Apply
reports.
soft Update Improvement Program


Products and Classifications
Products
Classifications
You can specify the products for which you want to synchronize updates.
Products:
ASP.NET Web Frameworks
Azure Connected Machine Agent
Azure Connected Machine Agent 3
Azure File Sync
Azure File Sync agent updates for Windows Server 2012 R2
Azure File Sync agent updates for Windows Server 2016
Azure File Sync agent updates for Windows Server 2019
Azure File Sync agent updates for Windows Server 2022
Azure File Sync agent updates for Windows Server 2025
Azure IoT Edge for Linux on Windows
Azure IoT Edge for Linux on Windows Category
Bing
Bing Bar
Azure File Sync agent updates for Windows Server 2025
OK
Cancel
Apply

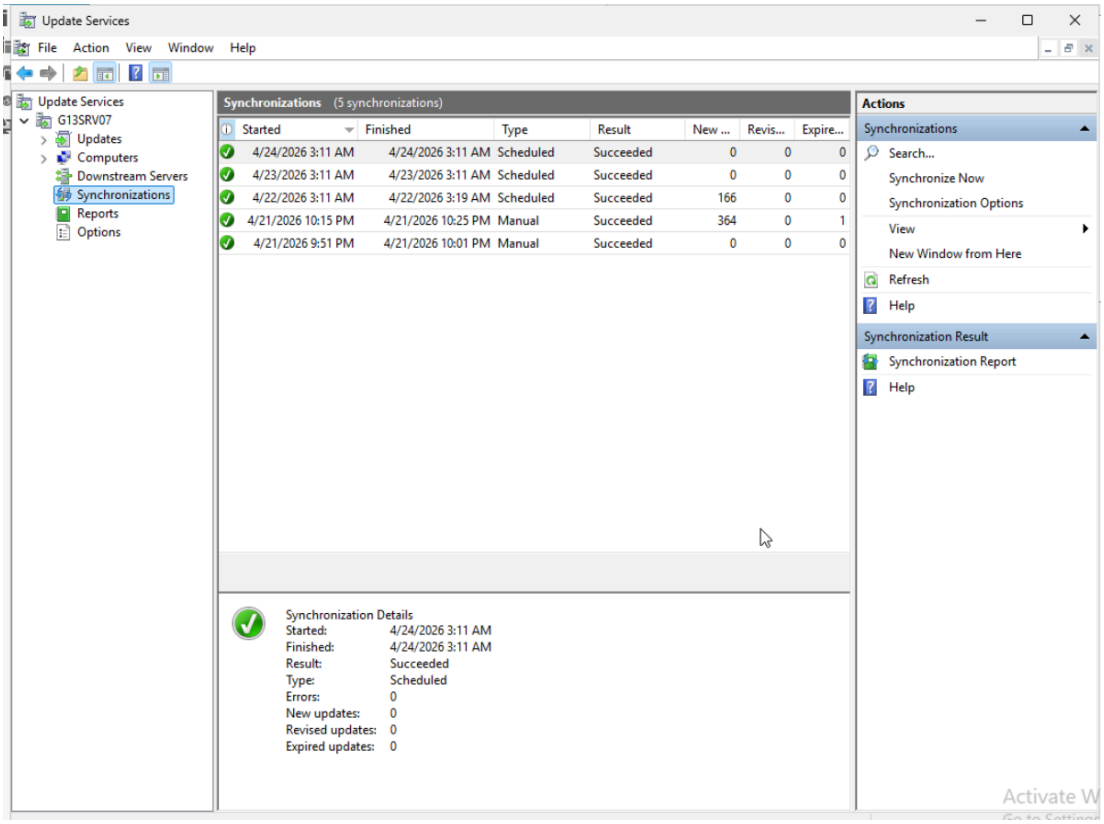


Figure 7: Lifecycle management showing synchronization success, storage optimization policies, and product-specific targeting.

The update repository is optimized to preserve server resources. By configuring a Post-Install Content Directory on the D: drive via `wsusutil.exe` (executing from `C:\Program Files\Update Services\Tools`), the system volume is protected from storage exhaustion. The synchronization history confirms a consistent track record of success, while product selections are granularly tailored to include critical components such as the Azure File Sync agent.

10. Strategic Conclusion

This project demonstrates a comprehensive mastery of enterprise-scale Windows infrastructure. By architecting a unified system centered on Active Directory, DFS resilience, and WSUS compliance, a production-ready environment was created that prioritizes security through granular delegation and operational reliability through remote automation. The successful integration of these technologies ensures that the `group13.c24200.cit.lcl` domain remains stable, scalable, and audit-ready for enterprise deployment.